

Mobility Axiom — Configuration Manual

Audience: IT staff, super admins, installers.

Pair with: **Quick Guide** (install) · **User Manual** (daily operations)

This manual covers **server setup**, **BWC camera registration**, **operators**, **storage**, **firewall**, and **video conference** configuration — with numbered steps.

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1. Before you start

Item	Requirement
Server PC	Windows 10/11 64-bit, LAN IP static recommended
Mobility pack	Unzipped; <code>Install-Mobility.bat</code> completed once
Docker Desktop	Installed and running (Video Conference only)
BWC cameras	Powered on, SIM/data or Wi-Fi, SIP screen accessible

Item	Requirement
Network	Server and BWCs on routable LAN; IPv4 only on camera keypad

Do not install separate Node.js — the pack includes `Mobility-Axiom\tools\node\`.

2. Open Server Config

1. Sign in as **super admin** (trial: `global`).
2. Click **Settings** tab.
3. Left panel → **Server Config**.
4. You see left nav: **Network & deployment** | **BWCs** | **Map groups** | **Dashboard Auth**.

Operator accounts: Server Config is **read-only** — ask super admin for changes.

Read-only banner: Yellow bar explains you cannot save.

3. Network & deployment tab

Inside **Network & deployment**, section nav on the right:

3.1 Deployment

1. **Deployment mode:** choose **LAN server** for on-prem trial.
2. **Tenant / customer name:** optional label on reports.
3. Scroll to bottom → **Save server settings**.

3.2 LAN network (critical for BWC online)

1. Open **LAN network** section.
2. Set **BWC SIP server IP** = this PC's **IPv4** (e.g. `192.168.1.10`) — **not a hostname**.
3. Note **HTTP port** (default `3888`) and **video WebSocket port** (default `3889`).
4. **Save server settings**.

3.3 BWC camera register (SIP)

1. Open **BWC camera register** section.
2. Set **Platform ID**, **Realm**, **Password** — defaults work for trial; change for production.

3. Copy values from **Type on BWC** block — you will type these on each camera SIP screen.

3.4 Protocol

Same SIP password and realm for every BWC. Do not mix per-device passwords unless advanced setup.

3.5 Operator access

Set URLs shown to operators; optional HTTPS notes for production.

3.6 Site timezone

Set correct timezone for logs and evidence timestamps.

3.7 Storage link

Shortcut to evidence path settings (see §7).

4. Register BWCs

Tab: BWCs (left nav in Server Config).

4.1 Add each camera

For every physical BWC:

1. Click **Add row** (or empty row).
2. **Device ID** — exact ID from camera label / SIP menu (e.g. `34020000001320000001`).
3. **Officer** — nickname shown on map and fleet table.
4. **Map group** — optional colour group for pins and PTT.
5. Click **Save BWC list** at bottom.

4.2 Type on BWC (camera SIP screen)

On each camera's SIP configuration menu, enter:

Camera field	Enter this
SIP server / registrar	Server IPv4 from Network tab

Camera field	Enter this
Port	5060 (default)
Platform ID	From Server Config
Realm	From Server Config
Password	From Server Config
Device ID	Same as row in BWC list

Rules:

- Use **numbers and dots only** for server address — no DNS names on camera keypad.
- Device ID on camera **must match** Mobility list exactly.

4.3 Verify online

1. Save BWC list → wait ~30 seconds.
2. **Operations** → Device Summary → **Online**.
3. If Offline → see §13 troubleshooting.

4.4 Trial empty fleet

Pack sets `FM_SEED_BWC_ID=` empty — no demo device. You start with zero until you add rows.

5. Map groups

Tab: Map groups.

1. Create groups with name and colour (e.g. "Team A" = blue).
2. Assign each BWC row to a group in BWCs tab.
3. Used for: pin colour, video wall rotation, PTT groups.

6. Dashboard Auth

Tab: Dashboard Auth.

6.1 Create operator (read-only dispatcher)

1. Add row: username, password.
2. Role: **Operator** or custom toggles.
3. Enable: Operations, Evidence view, Command Wall, VC — disable Server Config edit.
4. Click **Save** on that row.

6.2 Create additional super admin

1. Add row with **Super admin** role.
2. Save row.

6.3 Permission toggles (custom)

Per-user checkboxes control each tab and export capabilities. Always click **Save** on the row after changes.

7. Evidence storage

Paths: Server Config → Storage sections, or **Evidence** → **Storage** sub-tab.

Path	Purpose
FTP upload folder	Docking station uploads
Live capture folder	Server-side live recordings
Evidence index	SQLite catalog on server (metadata)

Steps:

1. Browse or type folder on server disk (e.g. `D:\MobilityEvidence\ftp`).
2. **Save storage.**
3. **Scan FTP for evidence** — indexes files into Library.

NAS/SAN: IT must mount volume on Windows first; then point Mobility at mount letter.

SOS auto-record: Enable **Auto-record server video on SOS alarm** in evidence settings if required.

8. Video Conference

8.1 Docker / LiveKit

1. Install Docker Desktop.
2. Run `Install-Mobility.bat` — starts LiveKit container.
3. Verify: Video Conference tab connects.

8.2 VC Settings (in app)

Video Conference → **Settings** sub-tab:

- **Phone / browser video URL** — WebSocket URL phones use (e.g. `ws://192.168.1.10:7880`).
- Set to server LAN IP if cellular clients cannot reach `127.0.0.1` .

8.3 Android app

Distribute `MobilityConference-1.5.6.apk` — server URL = dashboard URL.

9. Centre Summary AI

- Model file bundled: `Mobility-Axiom\vendor\llm\` (~2 GB).
- No download on site (`FM_LLM_AUTO_DOWNLOAD=0`).
- First **Ask** may take 1–2 minutes to load model into RAM.
- Super admin only.

10. Firewall & ports

Allow on server host and perimeter:

Port	Protocol	Service
3888	TCP	Dashboard HTTP
3889	TCP	Live video WebSocket
3890	TCP	Audio WebSocket
5060	UDP/TCP	SIP (BWC registration)
7880	TCP	LiveKit HTTP

Port	Protocol	Service
7881	TCP	LiveKit RTC
40130–40200	UDP	RTP media (range may vary)
21	TCP	FTP dock (if enabled)
20000–20100	TCP	FTP passive (default range)

11. Trial license

Trial pack includes `storage/platform-license.json` :

- **5 BWC · 10 dashboard users · 1 year** (per issued license).
- Exceeding limits blocks new registrations — contact vendor to extend.

12. Passwords & audit

Change super admin password

Settings → Server Config → Dashboard Auth → your row → new password → **Save**.

Audit Trail

Settings → **Audit Trail** — filter exports for compliance.

13. Verification checklist

Step	Pass?
<code>Install-Mobility.bat</code> completed without error	☐
Docker running (whale icon steady)	☐
Dashboard opens <code>http://localhost:3888</code>	☐
Server Config → LAN IP = PC IPv4	☐

Step	Pass?
Each BWC row added and saved	☐
Camera SIP matches Type on BWC	☐
Fleet shows Online within 60s	☐
Pin shows live video	☐
PTT heard on BWC	☐
Evidence FTP path saved + scan works	☐
VC Join Room works in browser	☐
Operator test account login read-only	☐

If any step fails, note exact error and check User Manual §18 + server logs (`VIEW-LOG.bat` in Mobility-Axiom folder).